
PO12Q - Introduction to Quantitative Political Analysis II

Resit/Deferral Exam 2025/26

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Paper Code: PO12Q –
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Exam Rubric

Time Allowed: 2 Hours
Exam Type: Written - Standard
Approved Calculators: Permitted

Answer all questions in the answer booklet provided

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Instructions

- Part I contains the questions for the exam
 - These are divided into four blocks: A, B, C, and D.
 - Block A is compulsory and must be answered by all candidates.
 - You must then select ONE block out of B, C, and D.
 - Should you change your mind during the exam, indicate clearly which block you wish to have marked.
 - You can achieve a maximum of 120 points.
 - Information on the maximum number of points obtainable is outlined at the beginning of each block
- Part II contains the case study to which all questions in Block A of Part I refer
- Part III contains the relevant statistical tables
 - Normal curve tail probabilities are provided in Table 6, page 14.
 - t-distribution critical values are provided in Table 7, page 15.
 - χ^2 -distribution critical values are provided in Table 8, page 16.
- Part IV contains the formula collection and notation of the module
 - If you use notation other than in the formula collection (page 17) or notation table (page 19), give an explanation.
- Allowed time: 2 hours.
- A university-approved calculator is allowed.
- Round all intermediary and final results to three decimals.

Part I: Exam Questions

1 | Block A

80 points, points for each question in parentheses

All questions in this section relate to the case study in Part II.

Level 1 (each question 4 points)

1. Subsections 12.1 and 12.2: State the null- and alternative hypotheses underpinning Model 3.
2. Table 4: What is the observed t-value for the “Lived Poverty Index” coefficient in Model 1?

Level 2 (each question 6 points)

3. Table 4: Interpret the intercept in Model 1.
4. Table 4: Interpret the coefficient of “Secondary Education (%)” in Model 3.
5. Table 4: Interpret the model fit for Model 4.
6. Which of the regression models shown in Table 4 is the best model overall? Explain your reasoning.
7. Which of the three independent variables used to predict `trust` in the regression models in Table 4 has the best and which one has the worst predictive strength? Explain your answer.
8. Table 4: Why is the intercept in Model 4 different from the intercepts in Models 1 and 2?
9. Given the output in Subsection 13.2, is it required to run the following command? Why/why not?

```
coefTest(model4, vcov = vcovHC(model4, type="HC3"))
```

10. Subsection 13.1: Is (multi-)collinearity an issue in Model 7?

Level 3 (each question 8 points)

11. Consider the `modelsummary` code in Subsection 14. In conjunction with the previous code chunks in this case study, why would this code not produce Table 4? Explain your answer.
12. Section 11: Assess whether there is a statistically significant association between `povcat` and `educat`.
13. Compare Model 8 in Table 5 to Model 1 in Table 4. How do these models differ, which is the better specification, and why?

Bonus (4 points, you can achieve a maximum of 180 points)

- In which month does Flo’s cat Darwin celebrate his birthday?

40 points, 5 points for each statement

State whether each of the following statements is true or false. Explain why the statement is true or false.

1. With increasing degrees of freedom, higher values for the χ^2 test statistic are required to reject the null hypothesis.
2. The conditional expectation $E(y_i | x_i = 10)$ tells us the average value of y across all observations in the population.
3. If R^2 in a bivariate regression model is close to zero, we can conclude that there is no relationship whatsoever between the independent and the dependent variable.
4. If the sample variances in a two-sample test for a mean differ, we use Welch's Test.
5. The sampling distribution of the OLS estimator $\hat{\beta}$ is approximately normal when the sample size is smaller than 30.
6. To satisfy the Classical Linear Assumptions, all independent variables must be normally distributed.
7. When no correlation between X and ϵ exists, then the conditional mean of error terms is automatically zero.
8. The difference between the means in a two-sample test is equal to the slope coefficient in a bivariate regression model in which the grouping variable is used as the independent variable, and the continuous variable as the dependent variable.

40 points, points for each question in parentheses

Table 1 shows the mean institutional trust score (0–100; higher = more trust) and the percentage who reported voting in last national election for a sample of five countries from the case study in Part ?? . See Table 2 for full codebook. All intermediary results must be carried forward with three decimals, and final results must also be reported with three decimals.

1. Specify the null and alternative hypothesis for the relationship between the mean institutional trust score and the percentage who reported voting in last national election. **(4)**
2. Fit a linear regression model of the type $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$ to the data in Table 1. Specify the full sample regression function. **(18)**
3. Calculate and interpret the model fit. **(18)**

Country	Mean institutional trust	Percentage who reported voting
Nigeria	19.5	66.1
Kenya	46.5	56.2
South Africa	35.2	77.9
Uganda	52.0	64.3
Botswana	21.6	73.1

Table 1: Regression Data

4 | Block D

40 points, 8 points for each method / statement

Explain each of the following methods or concepts in your own words, and illustrate your explanation with a worked example of your choice.

1. Conditional Expectation Function
2. Ceteris Paribus
3. Polynomial
4. Hat-Values
5. Interaction Effect

Part II: Case Study

5 | Preface

The data set is an excerpt from the Afrobarometer Round 9: The Afrobarometer is a pan-African, nonpartisan survey research network that conducts public attitude surveys on democracy, governance, the economy, and society. Round 9 covers 39 African countries and was conducted between 2021 and 2023. The variables in this data set are country-level aggregates of individual-level survey responses. (Afrobarometer Project, 2024)

The data in the data set is for 2022.

6 | Code Book

variable	label
country	Country Name
trust	Mean institutional trust score (0–100; higher = more trust)
poverty	Lived Poverty Index (country mean; 0 = no deprivation, 1 = extreme)
corruption	Percentage perceiving widespread corruption among officials
education	Percentage with secondary education or above
turnout	Percentage who reported voting in last national election

Table 2: Variables in the Afrobarometer_PO91Q data set for 2022

7 | Preliminaries

```
afro <- read.csv("afrobarometer.csv")  
  
library(tidyverse)
```

8 | Descriptives for afro Data Frame

Table 3: Descriptive Statistics for afro Data Frame

	N	Mean	SD	Min	P25	Median	P75	Max
population	625	36 473.00	0.00	36 473.00	36 473.00	36 473.00	36 473.00	36 473.00
urban	625	49.96	25.07	5.00	28.10	51.20	70.90	94.80
age	625	26.90	3.95	18.00	24.50	26.90	29.40	39.10
education	607	28.50	13.87	2.00	18.80	28.30	37.60	68.40
poverty	625	0.47	0.23	0.05	0.27	0.47	0.67	0.85
corruption	607	54.38	17.86	8.00	42.25	53.90	67.20	95.00
trust	625	42.18	12.43	9.60	33.90	42.40	50.60	75.70
turnout	613	61.75	15.66	15.60	51.70	62.10	72.20	98.00
gdp_pc	619	2439.60	1979.10	300.00	1215.50	1950.00	2375.50	7862.00

9 | Proportion of missing values per variable

```
## district country population urban age education poverty
## 0.0000 0.0000 0.0000 0.0000 0.0000 0.0288 0.0000
## corruption trust turnout gdp_pc
## 0.0288 0.0000 0.0192 0.0096
```

10 | Recoding

10.1 Secondary Education

```
afro <- afro %>%
  mutate(educat =
    ordered(
      cut(education, breaks = c(0, quantile(afro$education, probs = 0.5, na.rm=TRUE), 100),
        labels = c("low", "high"))))
```

10.2 Lived Poverty Index

```
afro <- afro %>%
  mutate(povcat =
    ordered(
      cut(poverty, breaks = c(0, mean(afro$poverty), 1),
        labels = c("low", "high"))))
```

11 | Crosstabulations

```
table1 <- with(afro, table(povcat, educat))
table1
##          educat
## povcat low high
##    low  157  137
##    high 149  164
```

12 | Regression

12.1 Regression Models

```
model1 <- lm(trust ~ poverty, data = afro)

model2 <- lm(trust ~ corruption, data = afro)

model3 <- lm(trust ~ education, data = afro)

model4 <- lm(trust ~ poverty + corruption, data = afro)

model5 <- lm(trust ~ poverty + education, data = afro)

model6 <- lm(trust ~ corruption + education, data = afro)

model7 <- lm(trust ~ poverty + corruption + education, data = afro)

afro$log.trust <- log(afro$trust + 1)

model8 <- lm(log.trust ~ poverty, data = afro)
```

12.2 Results Tables

Table 4: Regression Models I

	Dependent Variable: Mean Institutional Trust (0–100)						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Lived Poverty Index	-28.347*** (1.824)			-28.180*** (1.763)	-29.235*** (1.838)		-29.363*** (1.768)
Corruption Perception (%)		-0.197*** (0.027)		-0.188*** (0.023)		-0.196*** (0.028)	-0.192*** (0.023)
Secondary Education (%)			0.021 (0.036)		0.033 (0.031)	0.045 (0.036)	0.059* (0.030)
(Intercept)	55.402*** (0.950)	52.901*** (1.564)	41.632*** (1.153)	65.528*** (1.532)	55.008*** (1.283)	51.592*** (1.809)	64.765*** (1.691)
Num.Obs.	625	607	607	607	607	590	590
R2	0.279	0.079	0.001	0.353	0.296	0.078	0.373
R2 Adj.	0.278	0.077	-0.001	0.351	0.293	0.075	0.370

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table 5: Regression Models II

	Dependent Variable: Logged Mean Institutional Trust (0–100)
	(8)
Lived Poverty Index	-0.702*** (0.047)
(Intercept)	4.046*** (0.025)
Num.Obs.	625
R2	0.261
R2 Adj.	0.260

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

12.3 Regression Graphs



Figure 1: Visualisation of Regression Models

13 | Testing the Classical Linear Assumptions

13.1 `vif()`

```
vif(model7)
##      poverty corruption  education
##  1.0000913   1.008365   1.009065
```

13.2 `bptest()`

```
bptest(model4)
##
## studentized Breusch-Pagan test
##
## data: model4
## BP = 0.89892, df = 2, p-value = 0.638
```

14 | modelsummary

```
library(modelsummary)

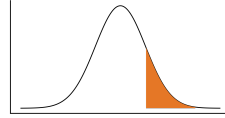
models <- list(
  "(1)" = model1,
  "(2)" = model2,
  "(3)" = model3,
  "(4)" = model4,
  "(5)" = model5,
  "(6)" = model6,
  "(7)" = model7,
)

cm <- c('poverty'      = "Lived Poverty Index",
       'corruption'    = "Corruption Perception (%)",
       'education'     = "Secondary Education (%)",
       '(Intercept)' = '(Intercept)')

modelsummary(models,
  gof_omit = 'DF|Deviance|Log.Lik|F|AIC|BIC|RMSE',
  stars = TRUE,
  coef_map = cm,
  group_tt(j = list("Dependent Variable: Mean Institutional Trust (0-100)" = 2:6))
```

Part III: Statistical Tables

15 | Normal-Distribution



z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641
0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002

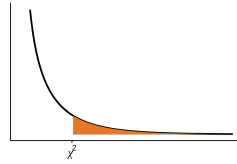
Table 6: Right-Tail Probabilities under the Normal Distribution

16 | t-Distribution

df	Confidence Level									
	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%
	One-Tail Probability									
	$t_{0.25}$	$t_{0.20}$	$t_{0.15}$	$t_{0.10}$	$t_{0.05}$	$t_{0.025}$	$t_{0.01}$	$t_{0.005}$	$t_{0.001}$	$t_{0.0005}$
1	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.3	636.6
2	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.33	31.60
3	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.22	12.92
4	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300
z	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291

Table 7: Probabilities under the t-Distribution

17 | χ^2 - Distribution



df	Right-Tail Probability								
	0.995	0.99	0.975	0.95	0.9	0.1	0.05	0.025	0.01
1	0.00	0.00	0.00	0.00	0.02	2.71	3.84	5.02	6.63
2	0.01	0.02	0.05	0.10	0.21	4.61	5.99	7.38	9.21
3	0.07	0.11	0.22	0.35	0.58	6.25	7.81	9.35	11.34
4	0.21	0.30	0.48	0.71	1.06	7.78	9.49	11.14	13.28
5	0.41	0.55	0.83	1.15	1.61	9.24	11.07	12.83	15.09
6	0.68	0.87	1.24	1.64	2.20	10.64	12.59	14.45	16.81
7	0.99	1.24	1.69	2.17	2.83	12.02	14.07	16.01	18.48
8	1.34	1.65	2.18	2.73	3.49	13.36	15.51	17.53	20.09
9	1.73	2.09	2.70	3.33	4.17	14.68	16.92	19.02	21.67
10	2.16	2.56	3.25	3.94	4.87	15.99	18.31	20.48	23.21
11	2.60	3.05	3.82	4.57	5.58	17.28	19.68	21.92	24.72
12	3.07	3.57	4.40	5.23	6.30	18.55	21.03	23.34	26.22
13	3.57	4.11	5.01	5.89	7.04	19.81	22.36	24.74	27.69
14	4.07	4.66	5.63	6.57	7.79	21.06	23.68	26.12	29.14
15	4.60	5.23	6.26	7.26	8.55	22.31	25.00	27.49	30.58
16	5.14	5.81	6.91	7.96	9.31	23.54	26.30	28.85	32.00
17	5.70	6.41	7.56	8.67	10.09	24.77	27.59	30.19	33.41
18	6.26	7.01	8.23	9.39	10.86	25.99	28.87	31.53	34.81
19	6.84	7.63	8.91	10.12	11.65	27.20	30.14	32.85	36.19
20	7.43	8.26	9.59	10.85	12.44	28.41	31.41	34.17	37.57
22	8.64	9.54	10.98	12.34	14.04	30.81	33.92	36.78	40.29
24	9.89	10.86	12.40	13.85	15.66	33.20	36.42	39.36	42.98
26	11.16	12.20	13.84	15.38	17.29	35.56	38.89	41.92	45.64
28	12.46	13.56	15.31	16.93	18.94	37.92	41.34	44.46	48.28
30	13.79	14.95	16.79	18.49	20.60	40.26	43.77	46.98	50.89
40	20.71	22.16	24.43	26.51	29.05	51.81	55.76	59.34	63.69
50	27.99	29.71	32.36	34.76	37.69	63.17	67.50	71.42	76.15
60	35.53	37.48	40.48	43.19	46.46	74.40	79.08	83.30	88.38
70	43.28	45.44	48.76	51.74	55.33	85.53	90.53	95.02	100.43
80	51.17	53.54	57.15	60.39	64.28	96.58	101.88	106.63	112.33
90	59.20	61.75	65.65	69.13	73.29	107.57	113.15	118.14	124.12
100	67.33	70.06	74.22	77.93	82.36	118.50	124.34	129.56	135.81

Table 8: Probabilities under the χ^2 -Distribution

Part IV: Formulae and Notation

18 | Formulae

Statistic	Formula
Crosstabulations	
χ^2 -Test Statistic	$\chi^2 = \sum \frac{(f_o - f_e)^2}{f_e}$
Two Sample Tests	
Standard Error of Difference	$se = \sqrt{(se_1)^2 + (se_2)^2}$
<i>Means</i>	
Standard Error of Difference	$se = \sqrt{\frac{se_1^2}{n_1} + \frac{se_2^2}{n_2}}$
Standard Error (pooled)	$se_0 = \sqrt{s_0^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$
Variance (pooled)	$s_0^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$
<i>Proportions</i>	
Standard Error	$se = \sqrt{\frac{\hat{\pi}(1 - \hat{\pi})}{n}}$
Standard Error of Difference	$se = \sqrt{\frac{\hat{\pi}_1(1 - \hat{\pi}_1)}{n_1} + \frac{\hat{\pi}_2(1 - \hat{\pi}_2)}{n_2}}$
Standard Error (pooled)	$se_0 = \sqrt{\hat{\pi}(1 - \hat{\pi}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$
Regression	
Adjusted R^2	$\bar{R}^2 = 1 - \frac{\sum \hat{\epsilon}_i^2 / (n - k - 1)}{\sum (y_i - \bar{y})^2 / (n - 1)}$
Coefficient of Determination	$R^2 = \frac{ESS}{TSS} = 1 - \frac{RSS}{TSS} = 1 - \frac{\sum \hat{\epsilon}_i^2}{\sum (y_i - \bar{y})^2}$
Estimated Variance	$\hat{\sigma}^2 = \frac{\sum \hat{\epsilon}_i^2}{n - p}$
Estimator of β_0	$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$
Estimator of β_1	$\hat{\beta}_1 = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^N (x_i - \bar{x})^2}$
Explained Sum of Squares	$\sum (\hat{y}_i - \bar{y})^2$
Residual Sum of Squares	$\sum \hat{\epsilon}_i^2 = \sum (y_i - \hat{y}_i)^2$
Estimated Standard Error of $\hat{\beta}_0$	$\hat{se}(\hat{\beta}_0) = \sqrt{\frac{\sum x_i^2}{n \sum (x_i - \bar{x})^2}} \hat{\sigma}$
Estimated Standard Error of $\hat{\beta}_1$	$\hat{se}(\hat{\beta}_1) = \frac{\hat{\sigma}}{\sqrt{\sum (x_i - \bar{x})^2}}$
Total Sum of Squares	$\sum (y_i - \bar{y})^2$

Table 9: Formulae for PO12Q

18.1 Matrices

$$\mathbf{X}'\mathbf{X} = \begin{bmatrix} 1 & 1 & \dots & 1 \\ x_1 & x_2 & \dots & x_n \end{bmatrix} \times \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} = \begin{bmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{bmatrix}$$

$$(\mathbf{X}'\mathbf{X})^{-1} = \frac{1}{n \sum x_i^2 - \sum x_i \sum x_i} \begin{bmatrix} \sum x_i^2 & -\sum x_i \\ -\sum x_i & n \end{bmatrix}$$

$$\mathbf{X}'\mathbf{Y} = \begin{bmatrix} 1 & 1 & \dots & 1 \\ x_1 & x_2 & \dots & x_n \end{bmatrix} \times \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} \sum y_i \\ \sum x_i y_i \end{bmatrix}$$

Symbol	Explanation
χ^2	Chi-Squared for test of independence
π	Population Proportion
$\hat{\pi}$	Sample Proportion
$\bar{\pi}$	Pooled Proportion
se_0	Standard Error under the Null Hypothesis
β	Regression Coefficient
$\hat{\beta}$	Estimated Regression Coefficient
ϵ	Error Term
$\hat{\epsilon}$	Estimated Error Term / Residual
A^{-1}	Inverse of Matrix A
A'	Transpose of Matrix A
I	Identity Matrix
R^2	R-Squared / Model Fit
σ^2	Mean Squared Error
k	Number of Slope Coefficients
$\bar{R}^2$	Adjusted R-Squared
$\log(x)$	Logarithm of variable x
α_i	Regression coefficients of secondary regression models
p	Total Number of Regression Coefficients, including the Intercept
VcV	Variance-Covariance Matrix
Ω	Is equal to $\sigma^2 I$, where σ^2 represents the mean squared error
t	Time period in time-series data

Table 10: Notation

Part V: References

List of References

Afrobarometer Project. (2024). Afrobarometer survey 2023, merged 39 country. <https://doi.org/10.2582/8/BS9K-ZY70>